

Coarse-threshold and coordinate refinement on anchored split frames

A conditional integer-multiplication saving of $5.2062354654791 \times 10^{-5}$

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With substantial OpenAI Codex assistance

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Abstract

We specify a finite refinement of the balanced, anchored split-frame construction. It combines an offset coarse-cell rule, compatible carry exchanges, and a cyclic coordinate flag at dimension 25. At dimensions 23 and 25 the selected words have 27,043 and 35,505 auxiliary roles. Under the inherited all-size framed-word, residual-compiler, analytic, routing, exact-recovery and fixed finite-alphabet multitape interfaces, the witness gives $T(n) = O(n(\log n)^{1-\kappa})$, with $\kappa = 52062354654791/10^{18}$. This is 0.0568021906% above the complete pinned PR82 profile. Newer community submissions report larger values; no latest-record or global-optimality claim is made. Exact finite checks and this written conditional argument do not formalize the complete multiplication theorem.

1 The selected graph and compiler choices

We retain the original-envelope geometry, rational $I + J$ basis, complete two-stage data network and complex branch of the cited construction chain. Address geometry is rational; bit payloads are over $\mathbb{F}_2$. The graph uses Avi Eisenberg’s interval strips and core-aware pair assembly [4], Rohan Garg’s split operation [2], and Chafik Boukhalfa’s anchored $[1, 1, 2]$ choices [9]. At every common point the lowest intact global pair is moved to the start of the inherited alternating point order. The first pair is split at the first two recursive levels and the middle group at the third, when it is a pair. Splitting is disabled at at most three remaining points. Thus the groups partition the points, have size one or two, and strictly reduce the recursive size.

For groups G_i, G_j , both evaluations

$$\sum_{b \in G_j} \left(\sum_{a \in G_i} e(a, b) \right) = \sum_{a \in G_i} \left(\sum_{b \in G_j} e(a, b) \right) \quad (1)$$

compute the same disjoint coarse edge sum. The balanced columns originate in PR69 [7]; the row and half-plane choices and future-compatible completion are from PR79 [12]. Our selected rule at $h = 23$ uses columns for $i + j < n_g + 1$ and rows otherwise, where indices start at zero and n_g is the current group count. This shifts the earlier boundary $i + j < n_g - 1$ by two. At $h = 25$, every cell uses rows. Support interning shares these intermediate sums with the strip computations. Every scalar output, disjoint addition and envelope is rechecked after construction.

Thomas DiFiore’s envelope ordering [10] bijectively rennumbers non-source nodes, preserves order within each envelope and remaps every argument, output and provenance reference. Derived support

caches are cleared after renumbering. Region order is **cover-core** at $h = 23$ and **reverse-node** at $h = 25$. The dependency checks remain enabled.

The compiler descends from eumemic’s reversible region synthesis [1], Chafik Boukhalfa’s reclamation [3], Rohan Arun’s profile-cost selection [5], and Dominik Scholz’s compatible live controls [6]. At $h = 23$ we retain descending-rank retired candidates and insertion-order live controls. At $h = 25$, both enter the temporary basis in profile-cost order. After desired output and retained-input rows, independent unit-completion rows are prioritized by their compatible future uses, then clean-signal support size and input index, as in PR79. These choices specify one finite algorithm; no optimal search is asserted.

2 Reclamation, exchanges and arbitrary dirty data

Let F_s be a carrier’s current frame, E the current region and G_s its scheduled next-use frame. Retired candidates satisfy $F_s \subseteq E$; a pending live control is eligible only if

$$F_s \subseteq E \subseteq G_s. \quad (2)$$

The temporary binary basis contains region anchors, compatible live controls and independent retired candidates. A dependent retired candidate supplies a literal relation $v_s = \sum_{a \in A} v_a$, with $s \notin A$. Emit $s \leftarrow s + a$ for all $a \in A$, raising both operands to E and recording every incidence. This clears the clean signal, not the arbitrary old dirty value. Each gate is invertible and each live control’s value is unchanged; (2) preserves its future eligibility.

For discovery, let $H(F, G)$ be the sum of $t \log t$ over the fixed-basis blocks of $F \rightarrow G$, evaluated on a rationally defined integer grid. A live control is scored by

$$H(F_s, G_s) - H(F_s, E) - H(E, G_s), \quad (3)$$

and a retired carrier uses the full endpoint instead of G_s . All operands of a proposed clearing expression are included. The score is a heuristic: later operations can change its prediction, and only the final complete physical profile is accepted.

The selected matching uses three passes of the improving single carry exchanges of PR70/76 [8, 11], supplemented by two-region cycles. The two independence constraints are the direct sum of the regions’ linear matroids and the unique-use partition constraint.

Proposition 1. *Suppose selected carries connect distinct regions g_1, g_2 to distinct uses u_1, u_2 . Replace them by existing eligible candidates (g_1, u_2) and (g_2, u_1) , provided each region’s desired output rows and remaining retained rows stay independent after its own replacement. Then cardinality, unique uses, causal eligibility and local invertibility are preserved.*

Proof. Each region loses one row and gains one row; the two stated rank tests are exactly its remaining linear constraint. The same two right uses remain occupied once each, and every new edge already belongs to the checked eligible set. Completing each independent row set to a square basis gives an invertible local map. Its literal row-XOR synthesis therefore works on arbitrary physical input values, including dirty components. $\square$

The selected words use 113 two-region cycles at $h = 23$ and 86 at $h = 25$. Open two-region reassignment paths were also screened; none was selected. These are finite choices, not a theorem of weighted-matroid optimality.

Let L be the resulting invertible scratch mixer, V the source injection and J the terminal scatter. The complete fresh-source map satisfies $JLV = I$. The chronological word

$$L, J, L^{-1}, V, L, J, L^{-1}, V \quad (4)$$

adds $JLd + JL(d + Vx) = x$ to the target over $\mathbb{F}_2$, restores arbitrary scratch d , and leaves the source unchanged. The reversed transposed word gives the opposite orientation. Both identities are independently replayed on every ordinary and dirty basis vector.

3 Coordinate flags and their exact costs

The coordinate-conjugacy method is inherited from PR74/78/82 [10, 13]. For a point permutation matrix P , $PJ = JP$, so P commutes with $I + J$ and preserves $I - J/9$. Consequently the frame projector obeys

$$P \Pi_{C,U} P^{-1} = \Pi_{PC,PU}. \quad (5)$$

Ranks, nondegeneracy, inclusions and the inherited denominator bounds are preserved. We relabel each source triple, output common point and triple, frame mask and scatter target consistently. Slots and literal mixer XORs are unchanged by this final relabeling. Thus the full source/target/dirty identity is conjugated, not assumed from a rank histogram.

At $h = 23$, the retained PR82 flag is specified by the coordinate order

$$(1, 3, 2, 4, 5, \dots, 19, 21, 20, 22, 0),$$

meaning that its j -th listed old coordinate maps to coordinate j . At $h = 25$, our selected flag is $i \mapsto i + 1 \pmod{25}$. The h23 discovery oracle retains the original coordinates; the h25 oracle uses the final cyclic flag for both carry and reclamation pricing. Acceptance in both cases uses the actual relabeled word and newly reconstructed profile.

Contiguous pivot blocks depend on coordinate order, so their costs need not be invariant under (5). They are regenerated. In contrast, the data component contains every pair of source triples. Independent point permutations induce a bijection of this complete Cartesian product, preserving the aggregate inherited data, endpoint and exterior classes, including all recovered exceptional cases. This would not justify retaining an arbitrary sample's profile.

Every scatter gate reads an auxiliary and writes a target, so the scatter gates commute. Conjugation can change their lexical target order. The verifier therefore checks the exact terminal-port multiset with multiplicities, while rejecting even canceling extra gates. The inherited frame schedule can still group retained-center reads and ordinary piece reads in its charged order.

4 The complete finite accounting

Every local profile includes actual operand raises, copied retained centers, ordinary output growth and final cleanup. With $v_h = \binom{h}{3}$ and auxiliary count R_h , its rank mass is

$$\sum_t t n_{h,t} = h R_h + h(h - 1). \quad (6)$$

The event log agrees with transitions independently reconstructed from the literal word. The unchanged fixed-basis profiler checks all ordered blocks with its bounded-minor/CRT rules, with zero modular disagreements.

Quantity	$h = 23$	$h = 25$
Sources v_h	1,771	2,300
Auxiliary roles R_h	27,043	35,505
Mixer XORs	167,165	233,668
Clearing XORs (included)	5,295	7,689
Wrapped length $4 L + 14v_h$	693,454	966,872
Ordinary and dirty basis columns	30,585	40,105
Local rank mass	622,495	888,225

Put $m = 575$, $N = \binom{23}{3}\binom{25}{3} = 4,073,300$. Replicate the local profiles 2,300 and 1,771 times respectively. The full counts are

$$\begin{aligned}
W &= 2N + 2300R_{23} + 1771R_{25} = 133,224,855, \\
L_0 &= 2300 \cdot 23 \cdot 22 + 1771 \cdot 25 \cdot 24 = 2,226,400, \\
s &= mW - N + L_0 = 76,602,444,725, \quad mW - s = 1,846,900.
\end{aligned} \tag{7}$$

Besides the actual local profiles, the complete list contains two data profiles $9[1] + [21] + [17] + [481]$ per data pair, one paid endpoint copy per pair, and every exterior and source-growth class. Each child width is positive and less than 575; the maximum is 529. All integer multiplicities are stored in `selected/certificate.json`.

5 Exact exponent and conditional theorem

For a bit saving $a > 0$, the complete characteristic is

$$F(a) = \frac{1}{mW} \sum_t n_t t \exp\left(a \log \frac{m}{t}\right). \tag{8}$$

The strict witness is

$$a = \frac{26032532642421}{5000000000000000000}. \tag{9}$$

Rational atanh-series bounds enclose every logarithm. For $0 \leq u \leq v < 1$, a degree-eight Taylor lower bound at u and the upper bound

$$\sum_{j=0}^8 \frac{v^j}{j!} + \frac{v^9}{9!(1-v/10)} \tag{10}$$

are used with outward rounding. Exact arithmetic proves an upper moment below one at (9) and a lower moment above one at $a + 10^{-18}$. A second evaluation uses logarithm range reduction by $3/2$, 60 atanh terms and a degree-twelve exponential bound and confirms both signs. The complete PR82 list, with its own width, has a lower moment above one at (9).

Theorem 1 (Conditional on the inherited interfaces). *Assume the base multiplication theorem and the all-size ordered residual compiler, framed-word transfer, fixed finite-alphabet multitape layout, analytic/semantic reductions, routing, eligible-prime setup and exact-recovery interfaces used by the cited chain. Then*

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{52062354654791}{10^{18}}.$$

Proof. Use (4)–(9) as the fixed bit interface in the inherited balanced assembly. The complex saving remains $717/10^7$, $\beta = 1/20$, and the semantic guard remains one. Set $\eta = 10^{-12}$, $q = a(1 - 2\eta)$, $\epsilon = (1 - \eta)/(1 + q)$. The controlling margin is $G = \epsilon q$. All 47 strict constraints and seven margins are recomputed, and $\kappa < G < \kappa + 10^{-18}$.

The actual width now requires 27 bits. The bit halving degree is nine, since $2 \cdot 529^9 < 575^9$ and $2 \cdot 529^8 \geq 575^8$. The unchanged complex branch has degree 20 and wire bound 30. Thus the row coefficient is $27 \cdot 9 + 30 \cdot 20 = 843$, and the inherited degree-2000 row stock has gap $2000 - (51/25)843 = 7007/25 > 0$, with suffix slope 8000. The complex literal scalar charge is unchanged. The semantic, row-stock, remaining strict inequalities and sufficient arithmetic cutoffs are checked with these derived fields. Positive final slack gives the inherited logarithmic absorption. Arithmetic cutoffs are not a complete operational threshold. $\square$

The comparison is to PR82 commit 3410b940aa26. The improvement is 0.0568021906%. Later community submissions have larger reported exponents; this is not a latest-record claim. Our earlier PR77 witness remains reproducible at 83298467291d.

6 Reproduction, attribution and limits

Run `make -C research/balanced-split-reclamation verify` for deterministic word regeneration, independent full dirty-basis replay in both orientations, actual transition/profile reconstruction, exact arithmetic and 17 focused failure-mode tests. These include negative indices, aliased sources, omitted paid children, wrong or canceling uncharged scatter ports, stale bridge fields, adjacent exponent-grid rejection and disabled Python assertions. Run the repository’s `make -j1 verify` for its retained broad checks. Run receipts record environmental interruptions and any sequential continuation separately. Source freezes are explicit and are never refreshed by verification.

The finite checks do not execute or formalize the entire asymptotic machine, prove global optimality, establish the inherited interfaces independently, or show practical speedup. This package and its additional checks used substantial OpenAI Codex assistance and are not independent human expert review. The new contribution is the explicit selected refinement and checked comparison; general carry exchanges, coordinate conjugacy and inherited graph/compiler techniques retain their authorship and notices.

References

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- [13] Rohan Arun. Envelope-balanced coordinate pricing, PR82, 3410b940aa26. <https://github.com/CrocSwap/integer-mult-bounds/pull/82>.